7.3.4 Successive Over-Relaxation (SOR) ¶

Recall that if $A = D - L - U$ where $-L$, D, and $-U$ are the strictly lower triangular, diagonal, and strictly upper triangular parts of A, then the Gauss-Seidel iteration for solving $Ax = y$ can be expressed as $x^{(k+1)} = (D - L)^{-1}(Ux + y)$ or, equivalently, $\chi_i^{(k+1)}$ solves

$$\sum_{j=0}^{i-1} \alpha_{i,j}\chi_j^{(k+1)} + \alpha_{i,i}\chi_i^{(k+1)} = -\sum_{j=i+1}^{n-1} \alpha_{i,j}\chi_j^{(k)} + \psi_i.$$

where any term involving a zero is skipped. We label this $\chi_i^{(k+1)}$ with $\chi_i^{\mathrm{GS}\,(i+1)}$ in our subsequent discussion.

What if we pick our next value a bit further:

$$\chi_i^{(k+1)} = \omega\chi_i^{\mathrm{GS}\,(k+1)} + (1-\omega)\chi_i^{(k)},$$

where $\omega \geq 1$. This is known as ***over-relaxation***. Then

$$\chi_i^{\mathrm{GS}\,(k+1)} = \frac{1}{\omega}\chi_i^{(k+1)} - \frac{1-\omega}{\omega}\chi_i^{(k)}$$

and

$$\sum_{j=0}^{i-1} \alpha_{i,j}\chi_j^{(k+1)} + \alpha_{i,i}\left[\frac{1}{\omega}\chi_i^{(k+1)} - \frac{1-\omega}{\omega}\chi_i^{(k)}\right] = -\sum_{j=i+1}^{n-1} \alpha_{i,j}\chi_j^{(k)} + \psi_i$$

or, equivalently,

$$\sum_{j=0}^{i-1} \alpha_{i,j}\chi_j^{(k+1)} + \frac{1}{\omega}\alpha_{i,i}\chi_i^{(k+1)} = \frac{1-\omega}{\omega}\alpha_{i,i}\chi_i^{(k)} - \sum_{j=i+1}^{n-1} \alpha_{i,j}\chi_j^{(k)} + \psi_i.$$

This is equivalent to splitting

$$A = \underbrace{\left(\tfrac{1}{\omega}D - L\right)}_{M} - \underbrace{\left(\tfrac{1-\omega}{\omega}D + U\right)}_{N},$$

an iteration known as successive over-relaxation (SOR). The idea now is that the relaxation parameter ω can often be chosen to improve (reduce) the spectral radius of $M^{-1}N$, thus accelerating convergence.

We continue with $A = D - L - U$, where $-L$, D, and $-U$ are the strictly lower triangular, diagonal, and strictly upper triangular parts of A. Building on SOR where

$$A = \underbrace{\left(\tfrac{1}{\omega}D - L\right)}_{M_F} - \underbrace{\left(\tfrac{1-\omega}{\omega}D + U\right)}_{N_F},$$

where the F stands for "Forward." Now, an alternative would be to compute the elements of x in reverse order, using the latest available values. This is equivalent to splitting

$$A = \underbrace{\left(\tfrac{1}{\omega}D - U\right)}_{M_R} - \underbrace{\left(\tfrac{1-\omega}{\omega}D + L\right)}_{N_R},$$

where the R stands for "Reverse." The symmetric successive over-relaxation (SSOR) iteration combines the "forward" SOR with a "reverse" SOR, much like the symmetric Gauss-Seidel does:

$$\begin{array}{rcl} x^{(k+\frac{1}{2})} & = & M_F^{-1}(N_F x^{(k)} + y) \\ x^{(k+1)} & = & M_R^{-1}(N_R x^{(k+\frac{1}{2})} + y). \end{array}$$

This can be expressed as splitting $A = M - N$. The details are a bit messy, and we will skip them.